Discussion on Active Fire Protection Systems for Offshore Substation of Wind Power Station

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Abstract. The offshore wind development activity is accelerating worldwide in recent years and the fire protection of the offshore substation gets more and more important accordingly. In this article, the commonly adopted active fire protection systems including portable extinguisher, water mist system, deluge system, pressure water-spraying system, sprinkler system, gaseous system, foam system recommended in three representative specifications or standards namely *Offshore Standard* (DNV-OS-J201) *Offshore Substations for Wind Farms* compiled by DET NORSKE VERITAS, *Code for 110 kV ~ 220 kV Offshore Substation Design of Wind Power Projects* (*NB/T 31115–2017*) compiled by China electricity council, *Standard for design of offshore wind farm* (*GB/T 51308–2019*) compiled by China renewable energy engineering institute, are listed, compared and analyzed. The preferred system for each areas in offshore substation of wind power station is proposed. Considering the fire-fighting performance, environmentally-friendly characteristics, maintenance ease and cost of use, the water mist system is recommended for main transformer room, diesel engine room, reactor room, switch gear room, GIS room and temporary refuge or shelter area. The fixed pressure water- spraying system is recommended for machinery spaces. The deluge system is recommended for rooms containing cylinders with compressed gas. The sprinkler system is recommended for accommodations. The water monitors are recommended adopted on helideck. This article could provide some references for similar project engineering designs.

Keywords: Offshore Substation · Offshore Wind Power Station · Active Fire Protection System · Engineering Design

1 Introduction

The offshore wind energy has been developed for a relatively long history in Europe. The total offshore wind installations in Europe passed 30 GW by the end of 2022, making up 47% of the global total offshore [1]. In the US, the total installed offshore wind power capacity could reach up to 29 GW before 2030 [2]. For China, the offshore wind power capacity is expected up to be 1500 GW in 2050 [3]. Global Wind Energy Council (GWEC) market intelligence expects that over 380 GW of new offshore wind capacity will be added over the next decade (2023–2032), bringing the global total offshore wind capacity to 447 GW by the end of 2032 [1]. Predictably, with the depletion of finite

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energy resource such as oil and coal, as well as the land resources, renewable, clean and land-saving energy sources such as offshore wind resource development is expected to be more and more important [4]. As the scale of wind farms continues to increase, the fire safety is the key factor that cannot be ignored during their design, construction, operation and maintenance. For example, the installed capacity of Rudong offshore wind farm reaches 4380 MW, while the water depth reaches 60 m in Ledong offshore wind farm [5]. For the fire safety requirements of offshore substation of wind power stations, *Offshore Standard (DNV-OS-J201) Offshore Substations for Wind Farms* compiled by DET NORSKE VERITAS provides relatively complete and comprehensive technical provisions and acceptance criteria. Subsequently, other countries like China have also prepared their own specifications or standards for offshore substation fire safety based on DNV-OS-J201, latest research achievements and construction experiences. In this paper, the active fire protection systems recommended in three representative specifications or standards are listed, compared and analyzed. The three specifications or standards are i. *Offshore Standard* (DNV-OS-J201) *Offshore Substations for Wind Farms* compiled by DET NORSKE VERITAS, ii. *Code for 110 kV ~ 220 kV Offshore Substation Design of Wind Power Projects* (NB/T 31115–2017) compiled by China electricity council and iii. *Standard for Design of Offshore Wind Farm* (GB/T 51308–2019) compiled by China renewable energy engineering institute. Through comparison and analysis, the preferred active fire protection system for each areas in offshore substation of wind power station has been proposed, in order to provide reference for the design of similar projects.

2 Configuration Requirements of Active Fire Protection Systems

2.1 Portable Extinguisher

There are detailed description for the configuration of portable extinguisher in all three specifications or standards. The capacity and location of the portable extinguishers have been specified in the three codes with subtle differences. In particular, fluid extinguisher can be adopted according to *DNV-OS-J201*, while this kind of extinguisher has not been mentioned in the codes *NB/T 31115–2017* and *GB/T 51308–2019*.

2.2 Water Mist System

The code *NB/T 31115–2017* has the most detailed provisions for water mist systems. It specifies the minimum operating pressure, minimum duration, maximum system response time, maximum filling time of water distribution pipes. And it also makes clear provisions on the location of the protective nozzles and the water mist nozzles.

In the *DNV-OS-J201*, the recommended minimum capacity of mist systems is also specified, which is 10 L min^{-1} m^{-2} to provide exposure protection to equipment within that area.

As for the code *GB/T 51308–2019*, it mainly specifies the minimum operating pressure of the water mist system and the location of the water mist nozzles.

2.3 Deluge System

There is no description of deluge system in the codes *NB/T 31115–2017* and *GB/T 51308–2019*.

According to the *DNV-OS-J201*, deluge system could be adopted in storage areas for compressed gas. The system should be designed to achieve local control and remote control. The deluge firewater distribution pipe-works and deluge valves setting are also specified in this code.

2.4 Pressure Water-Spraying System

The application areas of pressure water- spraying system are not listed in the codes *NB/T 31115–2017* and *GB/T 51308–2019*, thus the setting requirements of pressure water-spraying system specification are also not be mentioned in these two codes.

DNV-OS-J201 specifies that the minimum effective average distribution of water should be 5 L min^{-1} m^{-2} in the spaces such as machinery spaces. In order to improve the effectiveness and timely response of the system, the system pressures in standby and startup condition are also specified. The requirements of main power supply and backup power supply used to drive the pressure water-spraying system pumps have been emphasized. Furthermore, the requirements for prevention the blockage of the nozzle by impurities in the water have been put forward in *DNV-OS-J201*.

2.5 Sprinkler System

Similar to the pressure water-spraying system, there is no description for sprinkler system in the codes *NB/T 31115–2017* and *GB/T 51308–2019*.

In the *DNV-OS-J201*, the application of the system is limited to such areas such as accommodation, where slow fire growth ($\alpha = 0.003$ kW s^{-2}) is expected.

2.6 Gaseous Systems

The requirements of gaseous system in *DNV-OS-J201* are specific and detailed. The setting requirements of fire extinguishing medium conveying pipeline, openings requirements for gaseous system rooms, the alarm device setting and personnel evacuation, are specified in detail. The design of the pressure level of the container shall take into account the pressure code requirements as well as the highest possible ambient temperature. The fire extinguishing medium should be stored in a safe, accessible and well-ventilated room. The entrance of the room should be connected with the open deck and independent from the protected area. The enclosure structure of the room, including the doors and windows should be air-tight, and the door should be open outwards.

The requirements of *NB/T 31115–2017* and *GB/T 51308–2019* for gaseous systems are relatively briefer. They only stipulate the selection principles of fire extinguishing mediums. For example, The fire extinguishing mediums that can release toxic gases and those can cause secondary damage to equipment are forbidden.

2.7 Foam System

DNV-OS-J201 specifies the foam discharge rate, the quantity of foam-forming liquid and the maximum expansion ratio of the foam. The codes *NB/T 31115–2017* and *GB/T 51308–2019* however only specify the suitable application areas of this system such as the diesel engine room, as well as the the suitable type of the foam system namely the high-expansion foam extinguishing system.

2.8 Fire Water Pump System

For the configuration requirements of fire pumps including marine growth prevention in fire water system, the codes *NB/T 31115–2017* and *GB/T 51308–2019* are described relatively in principle, while the concrete measures such as installation of the inlet strainers of fire pumps have been put forward in the *DNV-OS-J201*.

3 Methodology

To scientifically and rationally select the appropriate fire extinguishing system for each functional room of an offshore substation in a wind power station, a methodological approach is employed. Firstly, candidate fire extinguishing systems are screened based on their compliance with relevant standard requirements. Secondly, a comprehensive evaluation of the screened systems is conducted with respect to multiple criteria, including effectiveness, environmental impact, and maintenance feasibility and cost. This multi-criteria analysis aims to identify the most suitable fire extinguishing system that balances performance, environmental sustainability, and economic feasibility. The specific selection technology pathway is illustrated in Fig. 1 below.

4 Analysis and Discussion

4.1 Active Fire Control Systems Recommendation in Three Representative Specifications or Standards

The recommended active fire protection systems of three relevant specifications or standards are shown in Table 1.

4.2 Fire Suppression Effectiveness

Water mist system, gaseous system and foam system can all be used to extinguish fires [6]. However, when considering the factor of preventing the fire from reigniting, the gaseous system should be excluded [7].

4.3 Environmental Impact

Water mist system is a new type of environmentally-friendly fixed extinguishing system which contains the effect of heat absorption and oxygen insulation [8]. The foam concentrate however contains surface active agent, which may be harmful to the marine environment after fire extinguishing use [9, 10].

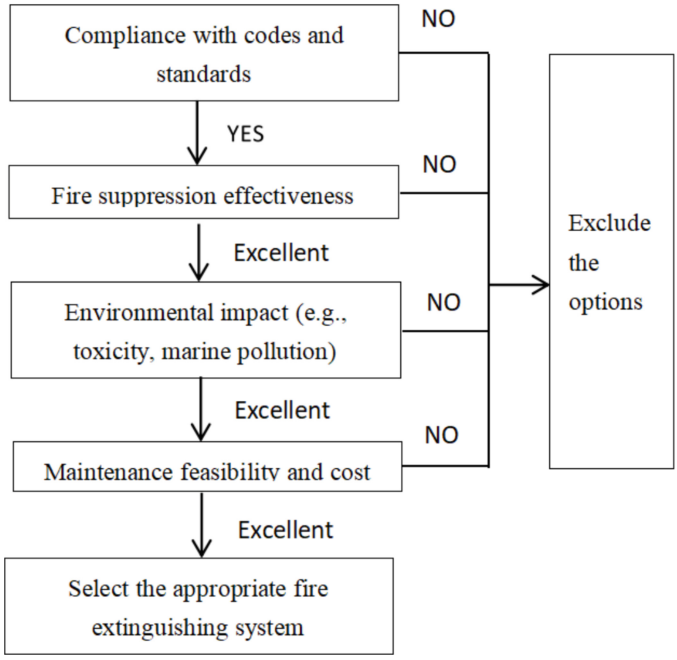

Fig. 1. Overview of method for selection of appropriate fire extinguishing system for each functional room of an offshore substation in a wind power station

4.4 Maintenance Feasibility and Cost

The foam concentrate used in foam fire extinguishing systems must be periodic replacement to ensure its fire-fighting performance. However, the offshore substations are generally far from land, so that the replacement and maintenance work of the foam concentrate are inconvenience. Moreover, considering the fire water source is usually seawater, seawater foam concentrate, which is more expensive compared to the conventional concentrate, has to been adopted.

4.5 Fire Extinguishing System Selection for Each Functional Room

4.5.1 Main Transformer Room

Water mist system, gaseous system and foam system can all be adopted for main transformer room fire control. For offshore substations with adequate area to set fire water tanks and fire pumps, gaseous system which is unable to prevent the fire from reigniting is not recommended. Although foam system is also effective for fire extinguishing, water mist system is more suitable for offshore substation of wind power projects considering the environmental impact.

Table 1. The recommended active fire protection systems

Areas	DNV-OS-J201	*NB/T 31115–2017*	*GB/T 51308–2019*
①All area	Portable fire extinguishers	Portable fire extinguishers	Portable fire extinguishers
②Main transformer room	Water mist system or gaseous system	Water mist system or foam system	Water mist system or foam system
③Diesel engine room, reactor room	Water mist system or gaseous system	Water mist system or foam system	Water mist system or foam system
④Switch gear room, GIS room	Water mist system or gaseous system	Water mist system or gaseous system	Water mist system or gaseous system
⑤Communication relay room, accumulator room, low voltage switchboard room	Water mist system or gaseous system	Water mist system or gaseous system	Water mist system or gaseous system
⑥Temporary refuge or shelter area	/	Water mist system	/
⑦Machinery spaces	Pressure water-spraying system or foam system	/	/
⑧Accommodation	Sprinkler system	/	/
⑨Rooms containing cylinders with compressed gas	Deluge system	/	/
⑩Helideck	Water monitors or foam system and dual-purpose nozzle and hoses	Water monitors or foam system	Foam and water dual water gun hydrant system

Note: / represents no description in the corresponding specifications or standards

4.5.2 Diesel Engine Room, Reactor Room, Switch Gear Room and GIS Room

The points of the fire extinguishing system selection for diesel engine room, reactor room, switch gear room and GIS room are similar to that of main transformer room. Therefore, the water mist system is the preferred extinguishing system for these areas.

4.5.3 Temporary Refuge or Shelter Area

Water mist system is recommended for temporary refuge or shelter area fire extinguishing and it is also the recommended fire extinguishing system in accommodation, public spaces and service areas on passenger ships by International Maritime Organization (IMO).

4.5.4 Machinery Spaces

Based on the DNV-OS-J201 and previous study [11], fixed pressure water-spraying system is suitable for extinguishing fires in machinery spaces, and it is also the recommended fire extinguishing system in Regulation 10 of Safety of Life at Sea (SOLAS). Considering the defects of foam system mentioned in part 3.2, fixed pressure water-spraying system is the preferred extinguishing system for machinery spaces.

4.5.5 Accommodation

Sprinkler system is widely used in cruise ship accommodation [12], transient accommodation buildings [13], student's accommodation [14] and residences [15]. This active fire protection system has a long record of property and life protection. In terms of reliability, safety and economy, sprinkler system is very suitable for fire protection for accommodations of offshore substations.

4.5.6 Rooms Containing Cylinders with Compressed Gas

Rooms containing cylinders with compressed gas has a relatively high risk of fire and explosion. Meanwhile, deluge system has been proved to been effective for fire protection for such places with high fire and explosion risk such as explosive processing building [16], LPG tanks [17] and even LPG vessels [18]. Hence, for rooms containing cylinders with compressed gas, deluge system is recommended for fire protection.

4.5.7 Helideck

The specified active fire protection systems for helicopter decks are basically consistent in the three specifications or standards. The relevant provisions are based on the requirements of the *Guide to helicopter/ship operations* published by International Chamber of Shipping (ICS).

According to the specifications and standards, water monitor system and foam system can both been adopted for fire fighting on helideck. For fire water monitor systems, they are widely installed on fire trucks or large space buildings as the fire fighting equipment and it's effectiveness for fire protection has been proved [19]. According to the previous study [7], fire monitor system is the most effective fire extinguishing method comparing to the foam fire extinguishing system (column type), water spray fire extinguishing system, foam fire extinguishing system (pipe network type) and water mist fire extinguishing system. Therefore, fire monitors are recommended for helideck fire protection. As for dual-purpose nozzle and hoses, they are used for the fire control by manual operation on the helideck.

5 Conclusions and Recommendations

In this paper, the selection of the appropriate fire extinguishing system for each functional room of an offshore substation in a wind power station was conducted through a hierarchical screening process, prioritized from the most critical to the least critical criteria. These criteria encompassed compliance with codes and standards, fire effectiveness,

environmental impact, and maintenance feasibility and cost. By systematically evaluating various fire extinguishing systems against these criteria, the optimal fire extinguishing system was identified and selected for each functional room of an offshore substation. The selection results are listed in Table 2 and each active fire protection systems should meet the requirements of corresponding specifications or standards.

Table 2. The optimal active fire protection systems for each functional room of an offshore substation

Areas	①	②	③	④	⑤	⑥	⑦	⑧	⑨	⑩
Optimal active fire protection systems	Portable fire extinguishers	Water mist system					Fixed pressure water-spraying system	Sprinkler system	Deluge system	**Fire monitors**

Note: The areas numbered ① to ⑩ referred to Table 1

Considering the huge development prospect of offshore wind power station in the future, the optimal design of the active fire protection systems for the offshore substations is worthy of further research and discussion.

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